Evaluating Classification Algorithms for Recognizing Figurative Expressions in Japanese Literary Texts

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Abstract. In this paper we introduce computational model for recognizing figurative expressions in Japanese language. As a part of the training data we use the set of almost 26,000 Japanese sentences comprising both similes and metaphors. These were collected manually from literary texts and hence constitute trustworthy and probably the largest existing resource of its kind. We use the data for classification task to evaluate its usability for figurativeness recognition. Precision score achieved by one of the classifiers utilized during the test shows that our model outperforms state-of-the-art methods in this aspect.

Keywords: Figurative language · Metaphor processing · Classification

1 Introduction

Enabling computers to recognize figurative use of language and making them understand its sense should be considered a necessary step in the process of further developing NLP systems and artificial intelligence in general. In many cases linguistic expressions can be understood both literally and figuratively depending on the context. To better understand the problem, try to imagine some ill-tempered person telling its humanoid robot to *get out of his face*. Robot would certainly become confused given it had not been taught the figurative sense of this expression. Example very similar to the one just presented is provided by Neuman in [10]. He rightly states that - although structurally very similar - the order *Give me the bottle* is drastically different from *Give me a break*. If the addressee was a robot, in order to make it understand what are we talking about, we would have to provide it with natural language understanding module that could differentiate between figurative and non-figurative senses.

Machine translation is another example of a field in which recognizing and understanding figurative senses of linguistic expressions becomes necessary. If one tries to translate *my sweet girl* into Japanese using probably the most popular automatic translation web service [16], he receives *watashi no amai onna no*

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ko as an output. This sounds unnatural and does not convey the illocutionary meaning (cf. [1] for elaborate description of illocution).

In the introduction to our recent work [2] we have briefly discussed the problem of metaphor-related terminology vagueness. Definitions of nearly all concepts associated with figurativeness available in most popular dictionaries seem to be highly ambiguous. Becoming acquainted with a vast number of theoretical works devoted to metaphor analysis does not resolve the problem: definitions of terms related to figurative language vary from author to author. Because of it we have decided to treat as figurative every linguistic unit, be it word or multiword expression (MWE), used in non-literal sense. It follows that not only metaphors (creative and conventional ones) but also metonymies, synecdoches, similes, personifications and their opposites, idioms, proverbs and alike: all of them should be considered objects of our investigation. Only this kind of generous approach seems to allow one to avoid contradictions during identifying figurative expressions.

One of the ultimate goals of our current study is to create a model performing with precision high enough to entrust it with the task of automatic figurative expressions recognition. Getting it close to 100% would grant us with the possibility of using online texts as an exhaustless resource of such utterances. Since insufficient amount of data is still one of the biggest problems for various machine learning algorithms to encounter, creating a large corpus of figurative expressions would become a great contribution for other researchers in the field as well.

In this paper Japanese words are transcribed into alphabet using Hepburn romanization; they are indicated with italics.

2 Related Work

Recent years have seen considerable increase in works related to computational processing of figurative language. Numerous papers related to metaphor identification has been published.

Bulat et al. [4] proposed metaphor identification method using semantic representations of 541 concepts built based on their most relevant attributes (for example *is large*, *has four legs*, *has fur*, *has hair*, *has hooves*, *is brown* for MOOSE). This database was previously constructed and presented by McRae et al. [9]. Classification experiment conducted by the authors showed that utilizing property norm-based semantic representations of this kind can yield even better results than popular dense semantic representations (F1 score of 77% vs. 73% & 75% vs. 73%). Authors used Support Vector Machines classifier.

Tsvetkov and colleagues [13] conducted classification experiment using dataset comprising English expressions of subject-verb-object (SVO) and adjective-noun (AN) syntactic patterns (F1 score of 79% and 85% for SVO and AN types respectively). They performed pivot translation using bilingual dictionaries and show that their model yields satisfying results also for Spanish (76%, 72%), Farsi (75%, 74%) and Russian (84%, 77%). Experiment was conducted

using Random Forest classifier. In [13] it is suggested that the model may serve as a tool for classification independently of the target language. Scores achieved by the algorithm in question are indeed very good but in our opinion it requires even deeper investigation in order to prove model's universality. The reason for that can be found in Dobrovol'skij and Piirainen's book [5]. Authors remind the reader of Whorfian concept of Standard Average European (SAE) according to which all languages of Europe (and that means also English, Russian and Spanish) belong to the same *sprachbund* (linguistic area) and show many common features with numerous very similar idiomatic expressions as their prominent example. Since semantically nearly identical idioms can be found also in Finnish, it can be proved that genetic affiliation or linguistic topology is of a very little importance when it comes to figurative language. On the other hand figurative expressions found in Japanese are often very different from their SAE's counterparts. It is because for a long time Japanese has been culturally isolated from Western influences that its phraseology reveals many unique cultural components. For example multiple figurative expressions containing words related to *kimono* - traditional Japanese garment - can be found. *Kare ni sode no shita o tsukau* 'offer him a bribe' includes the idiom *sode no shita* 'bribe' (lit. 'under the sleeve') evoking an image of *kimono*. What is peculiar about this piece of clothing is that it has long, wide sleeves allowing its user to hide something (also money or a gift) inside. Trying to translate this idiom into English without giving up on speaking figuratively one should say 'under the table' rather than 'under the sleeve'.

When it comes to research devoted to Japanese language processing, Hashimoto et al. [8] proposed lexical knowledge base for idioms recognition. In the introduction to this article we have already touched upon the problem, but Hashimoto et al. also point out that AI's inability to identify idioms often results in mistranslation. Japanese sentence *Kare wa mondai o kaiketsu ni hone o otta* ('He struggled to solve the problem') erroneously translated as 'He broke his bone to the resolution of a question' serve the authors as the example to illustrate the issue. In this case translation algorithm failed to recognize *hone o oru* 'to make efforts; to take pains' (lit. 'to break a bone') as being used figuratively. Another thing indicated by the authors as posing a problem in machine translation is transformation of fixed phrases, specifically phraseme's transformation occurring by e.g. adding another word in-between its constituents (eg. *kare wa yaku ni tatsu* 'he is helpful' vs. *kare wa yaku ni sugoku tatsu* 'he is very helpful'), its constituents' conjugation (e.g. *kare wa yaku ni tatta* 'he was helpful') and so forth. They point out that in many cases idioms do not allow morphological modifications (passivization, causativization, etc.) and therefore it follows that any change within their inner structure might be a hint for that they are being used literally; for example in case of *kare ga hone o orareta* 'he got his bone broken [by someone/something]' literal reading is possible while figurative is not. Unfortunately in this case such restriction does not hold for causativization, since *kare ni hone o oraseru* can be understood in both literal 'make him break a bone' and figurative 'make him struggle/make effort' senses. Authors divided idioms

into 3 categories: a) non-transformable and non-ambiguous (e.g. *mizu mo shitataru* 'extremely handsome'), b) transformable and non-ambiguous (e.g. *yaku ni tatsu* 'to be helpful'), c) transformable and ambiguous (e.g. *hone o oru* 'to make efforts'). Rule-based classification algorithm presented by the authors achieved F1 score of 80% for idioms belonging to the class c). The model sometimes failed to distinguish true senses of ambiguous expressions which interpretations depend on larger context.

In order to resolve aforementioned problems, few years later together with Kawahara, Hashimoto has presented results for improved version of his algorithm by conducting classification experiment using large corpus of sentences collected solely from the Internet [7]. Authors gathered 146 Japanese idioms, this time only those which senses are ambiguous (e.g. above-mentioned *hone o oru* fig. 'to struggle; to make effort', lit. 'to break a bone' or *goma o suru* fig. 'to flatter', lit. 'to crash sesame'). The model was improved by using both common word sense disambiguation features (e.g. hypernyms of words in the surrounding context, etc.) and idiom-specific features (those related to morphological constraints); the features were extracted using JUMAN and KNP parsers. Improved version of the classification algorithm achieved accuracy scores of 89.25% and 88.86% depending on the features used.

3 Datasets and Annotation

3.1 Datasets for Classification Experiment

Training data prepared for the classification experiment was comprising 25,947 sentences used figuratively and corresponding number of sentences with supposedly literal senses. *Supposedly*, because while no genre is completely free of figurative language, it is texts belonging to encyclopedic descriptions, news articles and formal discussions that are likely to be used in literal senses for the most part. First group was obtained from the digitalized dictionary of figurative Japanese expressions edited by Onai [11]. The second was collected using three different sources: Japanese Wikipedia dump [21], local assembly minutes [17] and articles from Livedoor News [19]. As for the dictionary of figurative expressions, our laboratory was provided with its abridged electronic version not available on the market, primarily comprising 25,979 items.

Our test data was initially composed of modest number of 50 sentences randomly taken from Japanese novels' texts freely available at Aozora Bunko digital library [15]. In the process of annotation described in greater detail in the following subsection, 41 sentences got labeled as literal and only 9 as metaphorical. In order to make these numbers equal, we have transferred another 32 sentences with non-literal senses from train- to test-set. Eventually we have used 51,894 sentences (25,947 figurative and 25,947 literal) for training and 82 sentences (41 figurative and 41 literal) for testing.

As a preliminary step towards conducting the experiment, we have parsed sentences from both sets with JUMAN [18], removed all ASCII characters (as this

format does not function as the encoding standard for Japanese text), punctuation marks, simile indicators and stopwords. Table 1 shows the full list of simile indicators as well as units we decided to consider as stopwords. The latter do not convey much meaning in isolation while being among 100 most frequent items in the dictionary.

Table 1. Simile indicators and stopwords removed during preprocessing

Simile indicators	Stopwords
yō, *mitai*, *ppoi*, *gotoku*, *gotoshi*, *gotoki*, *kurai*, *marude*, *sazo*, *samo*, *ikanimo*, *atakamo*, *hodo*	*suru*, *itasu*, *iru*, *oru*, *aru*, *naru*, *kore*, *sore*, *kono*, *sono*, *kara*, *koto*, *mono*, *nado*, *keredomo*, *demo*, *watashi*, *da*, *noda*, *nda*, *nani*, *gozaimasu*, *no*, *wo*, *ni*, *to*, *ga*, *wa*, *ka*, *o*, *go*, *de*, *mo*, *ka*, *sa*, *yo*, *he*, *nai*, *nu*, *ne*, *na*, *tai*, *tsu*, *rareru*, *reru*, *seru*, *ru*

3.2 Annotation

We have asked three Japanese native speakers to annotate 50 sentences taken randomly from Aozora Bunko digital library. They were instructed to not only decide whether a sentence have literal or non-literal sense but also - in case of the latter - to mark the expression used figuratively. Instruction was prepared in Japanese, and it can be translated into English as follows: *Read the following sets of three sentences one by one. Try to decide whether any figurative expression (metaphor, simile, etc.) is used in the middle (the second) sentence and choose the correct answer from the parenthesis. If you think that there is a figurative expression in use, indicate this word or phrase using an underline.*

As shown by Dobrzynska [6], evaluation of whether certain expression is used figuratively or literally is often context-dependent (consider for example *all men are animals* or *piece of thrash*). We have therefore provided annotators with adjacent sentences as well (the previous and the next to the target one). Odd number of annotators prevented possible problems with disagreement: sentences were labeled according to the majority's decision.

Let us briefly discuss some examples of labeling controversies. Expression *doko made mo* (lit. 'anywhere', 'wherever') from the sentence: *kenkyū wa doko made mo kenkyū de aru* 'research is a research (in every respect)' has been labeled as literal by all three annotators. To our understanding this expression should be considered as an example of underlying container metaphor.

Another example of expression in our opinion mistakenly labeled as being used in literal sense would be the following: *jikaku ga tsuyoku natta* '(one) became

more aware' (lit. 'awareness became strong'). This is an example of personification and thus figurative expression, since awareness is not a living entity and therefore cannot become strong in literal sense.

Longer sentences sometimes included more than one linguistic unit used figuratively. In the following example several expressions with figurative senses can be found: *Hitode wo karizu, fūfu dake de mise o kirimawashita node, yoru no jūji kara jūniji koro made no ichiban tatekomu jikan wa me no mawaru hodo isogashiku, shōben ni tatsu hima mo nakatta* 'Because me and my wife were running the store (lit. 'cutting the store around') without anyone's help (lit. 'not borrowing a hand'), at the rush hour from 10 o'clock to around 12 o'clock in the evening it was so busy my head was spinning (lit. 'busy to the extent eyes were spinning') and I didn't even have time to take a piss'. In such cases sentences tended to get labeled as metaphorical but it is difficult to estimate which of the expressions used led the algorithm to make a correct prediction.

4 Classification Experiment

We have extracted word count vectors of 32,231 dimensions for each of 51,894 items in the training set. This was also the number of unique words in the dictionary created from all words present in the sentences used for training. Three standard state-of-the-art classifiers, namely Naïve Bayes (multinomial), Support Vector Machines (Linear SVC) and Random Forest Classifier (100 trees) were used via Scikit-learn library [20] for Python.

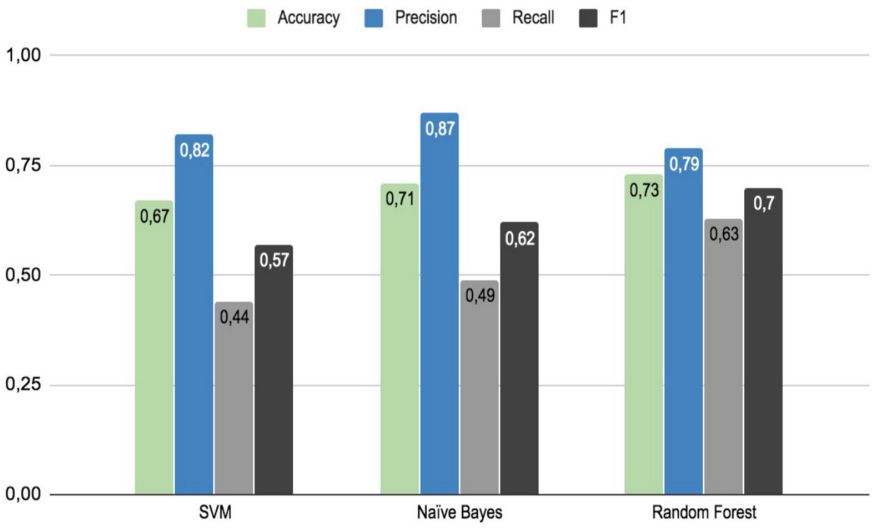


Fig. 1. Classification results

As can be noticed from Fig. 1, Random Forest has received the most balanced notes with F1 score of 70%. On the other hand, the highest precision has been

achieved by Naïve Bayes which leads to the conclusion that it might be the most suitable algorithm for the task of compiling figurative language corpus in Japanese.

5 Discussion and Future Work

One of the issues which we have encountered during the process of annotation is that native speakers sometimes fail to recognize figurative nature of certain expressions, especially those which we would classify as *conventional* (also *dead* or *frozen*) metaphors. It might be true that discerning figurative expressions from the literal ones becomes even more difficult in texts produced in one's native language. One might even risk a hypothesis that in order to get more accurate results it would be better to entrust the task to a non-native speaker with sufficient level of fluency in the target language. Pretheoretical tenet according to which figurative language is somewhat deviant, unusual or that it belongs solely to the literary genres like poetry seems to be still widespread which results in belief that we rarely speak metaphorically. On the other hand, in the process of foreign language acquisition it often becomes evident for the learners that semantically similar concepts are realized in different ways - often figuratively - depending on the language. Should annotators have educational background in linguistics or semiotics is yet another problem worth further discussion.

We need to somehow resolve this problem, because - as we have seen and as Tsvetkov and colleagues [13] point out - humans may disagree whether particular expression is used metaphorically or not. It might be therefore desirable to use some more objective guidelines like those presented by Steen et al. [12] rather than depend on annotators' intuition. It is also very difficult if not impossible to find large amount of data completely free of figurative expressions. As a source of literal data we have chosen to use encyclopedic descriptions, news articles and discussions that took place during local parliamentary sessions. This way of collecting literal data is obviously not perfect, but possibly the best as it can get without supervision.

As for the results achieved in the current experiment, in comparison with our last year's approach [3], scores of SVM and Naïve Bayes have improved significantly. Precision score of Naïve Bayes got nearly 30% better. The experiment has proven that, having it trained on large amount of data and with use of high-dimensional word vector representations, Naïve Bayes might be utilized as a powerful tool for figurative speech recognition. Encouraged by promising results achieved by Tsvetkov et al.'s model [14], in this trial we have utilized Random Forest classifier as well. It yielded the most balanced results out of all three algorithms used. We plan to continue on investigating methods for increasing classification model's performance, especially in regards to its precision. If we could get it close to 100% it would enable us to enlarge existing dictionaries of figurative expressions which in turn would become an invaluable contribution to NLP-related research. Some potential applications of well performing metaphor processing models can be found in the last chapter of [14].

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